Detecting fake news: binary classification of fake vs. true news based on textual characteristics of news articles

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Note: The model fitting code runs up to 5 minutes.

1 Summary

News is everywhere. In this digital era, there exists both true and fake news on the Internet, making it harder for readers and news agents to differentiate them. So, fake news detection is a technology with rising demands that helps defend information integrity. To solve this problem, applied machine learning is used for binary classification. The fitted Naive Bayes model with `MultinomialNB` as estimator and `title`, `text` and `subject` as features yields a test classification accuracy of 0.954, where the false positive and false negative rates are both low. The AP score and AUC value of the final model end up being 0.97 and 0.98 respectively, suggesting decent and satisfactory performance. Compared to the baseline dummy classifier, this Naive Bayes model classifies both true and fake news significantly better, which is reflected by consistently high and similar training, cross-validation and test scores. This machine learning model gives us confidence in its decent performance, which can potentially be deployed and used in the reality. But limitations do exist, since the collected data do not cover all possible types of news. So, when deployed in the wild, the model might perform badly in news of other realms, such as entertainment news. If more comprehensive data can be collected, then this model might handle more cases accurately.

2 Introduction

"Fake news", also known as disinformation, is deliberately misleading or false information presented as legitimate news (Service Canada 2024). Its rapid spread in recent years has becoming a pressing concern, particularly with the rise of social media platforms that accelerate the dissemination of content. These platforms are designed with the primary objective of capturing and retaining user attention for as long as possible. To that end, fake news often exploits this design to gain traction.

There is high public awareness of this issue in Canada. In 2023, Statistics Canada reported that 59% of Canadians admitted that they were "very or extremely concerned about online misinformation" in the Survey Series on People and their Communities (Bilodeau and Khalid 2024). In the United States, the concept of "fake news" grew in popularity during the 2016 presidential election. Allcott and Gentzkow (2017) estimated that false news stories were shared on Facebook at least 38 million times in the 3 months leading up to the election. Such widespread circulation of this false information can have serious consequences, including influencing public opinion and undermining trust in legitimate news sources. Thus, methods for detecting fake news may be beneficial in mitigating its spread.

This report investigates whether a machine learning model can accurately classify news articles as "fake" or "true" based on their textual characteristics, such as the article's title, subject and body content. Automated detection of fake news has the potential to mitigate the spread and impact of information, particularly in social media's fast-paced environment where information is shared instantly and at scale, from numerous sources.

3 Data

The dataset used in this analysis is the "Fake and Real News Dataset" available on Kaggle (Bisaillon 2020). It contains approximately 40,000 news articles published in the United States between 2015 and 2018. Each row in the dataset represents a news article, with a column for its title, subject, date, and body text.

The raw data from Kaggle was initially provided as two seperate datasets: one for the fake articles and one for the true articles. To prepare the data for modeling, we conducted some validation checks to make sure the data was in the expected format, with no missing observations or duplicate entries. The two datasets were then merged into one large one, with an additional column for the `target` value, indicating whether the article was labeled as "true" or "fake". To support model training and evaluation, we shuffled the rows and split the data into training (80%) and testing (20%) sets. Further validation was conducted on the training set to identify outliers and anomoulous correlations. Finally, we conducted exploratory data analysis (EDA) on the training data set to examine the distribution of the target variable and gain initial insights into the dataset.

	title	text
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Do

	title	text
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...

3.1 Data Validation

Part 1: Perform the following data validation checks: - Correct column names - No empty observations - Missingness not beyond expected threshold - Correct data types in each column - No duplicate observations

	title	text
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a cons
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender peopl
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counse
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign a
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - Preside
...	...	...
21412	'Fully committed' NATO backs new U.S. approach...	BRUSSELS (Reuters) - NATO allies on Tuesda
21413	LexisNexis withdrew two products from Chinese ...	LONDON (Reuters) - LexisNexis, a provider of
21414	Minsk cultural hub becomes haven from authorities	MINSK (Reuters) - In the shadow of disused So
21415	Vatican upbeat on possibility of Pope Francis ...	MOSCOW (Reuters) - Vatican Secretary of Sta
21416	Indonesia to buy $1.14 billion worth of Russia...	JAKARTA (Reuters) - Indonesia will buy 11 S

	title	text
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americ
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Dev
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwa
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announce
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day
...	...	...
22698	The White House and The Theatrics of 'Gun Cont...	21st Century Wire says All the world s a sta
22699	Activists or Terrorists? How Media Controls an...	Randy Johnson 21st Century WireThe majo
22700	BOILER ROOM – No Surrender, No Retreat, Heads ...	Tune in to the Alternate Current Radio Net
22701	Federal Showdown Looms in Oregon After BLM Abu...	21st Century Wire says A new front has just
22702	A Troubled King: Chicago's Rahm Emanuel Desper...	21st Century Wire says It s not that far awa

Part 2: Perform the following data validation checks on `true_df` and `fake_df`: - No outlier or anomalous values - Correct category levels (i.e., no string mismatches or single values)

	title	text
1442	Trump's Leaks To Russia May Well Have Put Ind...	Trump s ignorance, incompetence, and ego are p
10725	Republican Senator Kirk says he wants Senate t...	WASHINGTON (Reuters) - Republican Senator
2780	Massive copper mine tests Trump's push to slas...	SUPERIOR, Arizona (Reuters) - Rio Tinto's pr

	title	text
6773	Patton Oswalt To Bernie Fans: ‘You’re A F*cki...	Comedian Patton Oswalt has a knack for calling
13638	U.S. general sees no change in Pakistan behavi...	WASHINGTON (Reuters) - The top U.S. genera
...	...	...
7807	Clinton heavily favored to win Electoral Colle...	NEW YORK (Reuters) - After a brutal week fo
15556	Polish reforms of judiciary pose threat to rul...	BRUSSELS (Reuters) - Polish reforms of its jud
17920	Factbox: Japan main parties' key election pled...	TOKYO (Reuters) - Campaigning began on Tu
6819	Jared Fogle’s Pedophilic Texts Finally Releas...	Former Subway spokesman and outed pedophile
15905	Iran's Rouhani: Tehran-Moscow cooperation need...	ANKARA (Reuters) - Iran s President Hassan R

Part 3: Perform the following data validation checks on `complete_df`: - Target/response variable follows expected distribution - No anomalous correlations between target/response variable and features/explanatory variables - No anomalous correlations between features/explanatory variables

	title	text
1442	Trump’s Leaks To Russia May Well Have Put Ind...	Trump s ignorance, incompetence, and ego are p
10725	Republican Senator Kirk says he wants Senate t...	WASHINGTON (Reuters) - Republican Senator
2780	Massive copper mine tests Trump's push to slas...	SUPERIOR, Arizona (Reuters) - Rio Tinto’s pr
6773	Patton Oswalt To Bernie Fans: ‘You’re A F*cki...	Comedian Patton Oswalt has a knack for calling
13638	U.S. general sees no change in Pakistan behavi...	WASHINGTON (Reuters) - The top U.S. genera
...	...	...
7807	Clinton heavily favored to win Electoral Colle...	NEW YORK (Reuters) - After a brutal week fo
15556	Polish reforms of judiciary pose threat to rul...	BRUSSELS (Reuters) - Polish reforms of its jud
17920	Factbox: Japan main parties' key election pled...	TOKYO (Reuters) - Campaigning began on Tu
6819	Jared Fogle’s Pedophilic Texts Finally Releas...	Former Subway spokesman and outed pedophile
15905	Iran's Rouhani: Tehran-Moscow cooperation need...	ANKARA (Reuters) - Iran s President Hassan R

3.2 Data Preprocessing

4 Methods

4.1 EDA

```
<class 'pandas.core.frame.DataFrame'>
Index: 30903 entries, 1442 to 15905
Data columns (total 5 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
```

```
 0   title    30903 non-null  object
 1   text     30903 non-null  object
 2   subject  30903 non-null  object
 3   date     30903 non-null  object
 4   target   30903 non-null  object
dtypes: object(5)
memory usage: 1.4+ MB
```

As we can see from this summary, the training data contains 30903 observations. Each observation represents an article with a title, text, subject, date and target label. Each of these are non-null objects.

4.2 Count of Fake vs. Real News Articles

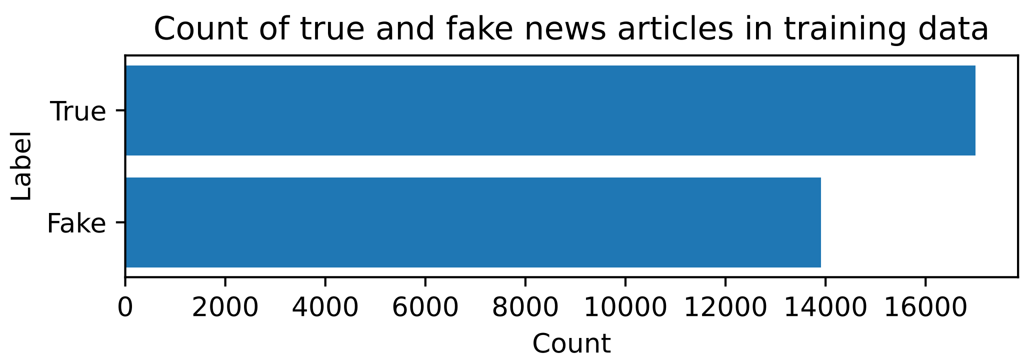

Figure 1: Count of true vs. fake news articles in training data set

From Figure 1, we can see that the training dataset is fairly balanced, with np.int64(13905) fake news articles and np.int64(16998) real news articles. This balance may help to prevent bias towards one class over the other when training our model.

4.3 Word Clouds for Titles Column

4.4 Classfication Modeling Analysis

4.5 Dummy Classifier

The dummy classifier acts as the baseline before building a more advanced model. The strategy used is “most_frequent” so that the dummy classifier always predicts the majority class. The goal later on is to build a classification model that does well compared to this dummy.

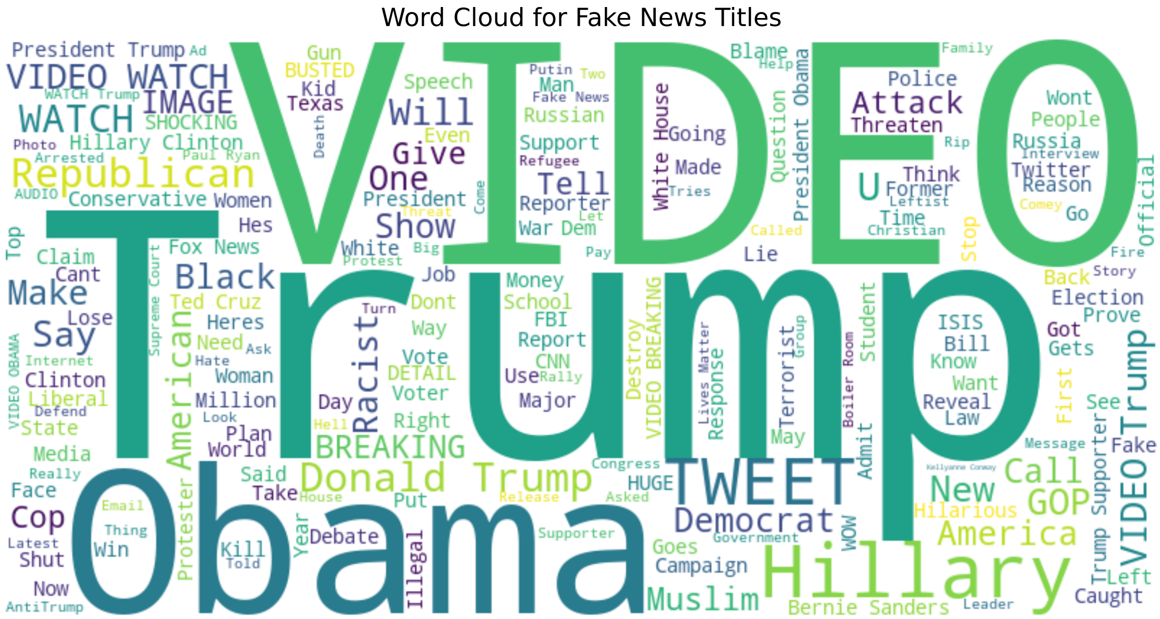

Figure 2: Word cloud for titles of fake news articles

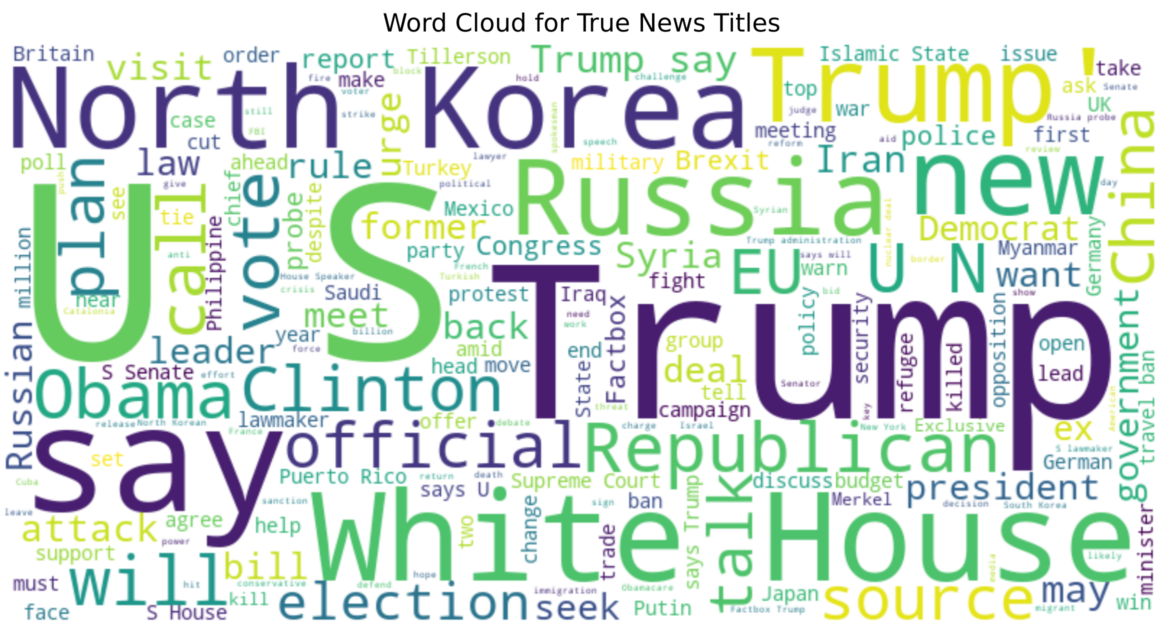

Figure 3: Word cloud for titles of true news articles

4.6 Naive Bayes Classifier

`MultinomialNB` is used as the estimator to deal with integer counts data, after proper preprocessing of the features. Two textual features (`title` and `text`) and a categorical feature `subject` with only two levels (political and non-political) are included in modelling. The `date` feature is dropped due to its irrelevance to the target. Genereally speaking, when the news is published is random information and has little to do with whether the news is fake or true.

As a first step, the features should be preprocessed. One-Hot encoding is applied to the categorical feature `subject` with `drop="if_binary"` to include only one column for this binary feature. For the two textual features, `title` and `text`, they are first transformed by `lambda x: np.ravel(x)` to be one-dimensional arrays. After being flattened, they are then passed to `CountVectorizer` to extract the words and number of occurrences from these two features. Note that `CountVectorizer` needs to be created twice, one for each textual feature.

Next, a machine learning pipeline `pipe` is created, with preprocessing and the Naive Bayes estimator.

Before fitting the model to the training set, hyperparameter tuning is carried out to boost classification performance. There are 3 hyperparameters being considered: `max_feautres` of the two `CountVectorizer`s, and `alpha` of `MultinomialNB`. The distributions of these hyperparameters to search from are specified below, followed by `RandomizedSearchCV` to do random search and cross validation.

Note that `RandomizedSearchCV` runs very slowly in this case. To reduce computation time, `n_iter` is set to only 10. Albeit doing only 10 searches, the best model found already performs very well, as we can see from the CV score and test scores below.

5 Results

As a baseline, dummy classifier is first fitted to get the test accuracy.

```
dummy_clf = DummyClassifier(strategy="most_frequent")
dummy_clf.fit(X_train, y_train)
dummy_pred = dummy_clf.predict(X_test)
dummy_accuracy = dummy_clf.score(X_test, y_test)
print("Baseline (Dummy) Accuracy: {:.3f}".format(dummy_accuracy))
```

```
Baseline (Dummy) Accuracy: 0.538
```

Using most frequent dummy classifier, the test score is low (only 0.538) due to a relatively balanced dataset. Now compare the Naive Bayes model with the dummy classifier.

```python
# Create preprocessing pipeline
title_feature = 'title'
text_feature = 'text'
categorical_feature = ['subject']
drop_feature = ['date']

title_pipe = make_pipeline(
    FunctionTransformer(lambda x: np.ravel(x)),
    CountVectorizer()
)
text_pipe = make_pipeline(
    FunctionTransformer(lambda x: np.ravel(x)),
    CountVectorizer()
)

preprocessor = ColumnTransformer(
    [
        ("one_hot", OneHotEncoder(drop="if_binary"), categorical_feature),
        ("title_vectorizer", title_pipe, "title"),
        ("text_vectorizer", text_pipe, "text"),
        ("drop", "drop", drop_feature)
    ]
)

pipe = make_pipeline(preprocessor, MultinomialNB())

param_dist = {
    "columntransformer__title_vectorizer__countvectorizer__max_features": randint(1, 200),
    "columntransformer__text_vectorizer__countvectorizer__max_features": randint(1, 5000),
    "multinomialnb__alpha": 10.0 ** np.arange(-7, 1)
}

random_search = RandomizedSearchCV(pipe,
                                   param_distributions=param_dist,
                                   n_iter=10, n_jobs=-1,
                                   return_train_score=True,
                                   random_state=123)

random_search.fit(X_train, y_train)
random_search.best_params_
train_score = random_search.score(X_train, y_train)
best_cv_score = random_search.best_score_
```

```
test_score = random_search.score(X_test, y_test)
```

After fitting the Naive Bayes model to the training set, the best hyperparameters and CV score can be extracted from the model. The best values of all 3 hyperparameters are in the middle of the search ranges instead of being on the edge, so the search ranges are specified appropriately. Under this model, the training score is 0.955. The best CV score is 0.954, and the test score is 0.954. Since the train, CV and test score are close to one another and are both high, underfitting or overfitting is not a concern here. Compared to dummy classifier, this Naive Bayes model performs a lot better in terms of prediction accuracy, increasing the test score from 0.538 to 0.954.

The Naive Bayes model demonstrates strong performance in distinguishing between fake and real news articles based on their textual characteristics, as shown in the word clouds (Figure 2 and Figure 3) which reveal distinct vocabulary patterns between fake and real news.

5.1 Model Evaluation

Below are the visualizations of the classification results on the test set, as a model performance assessment.

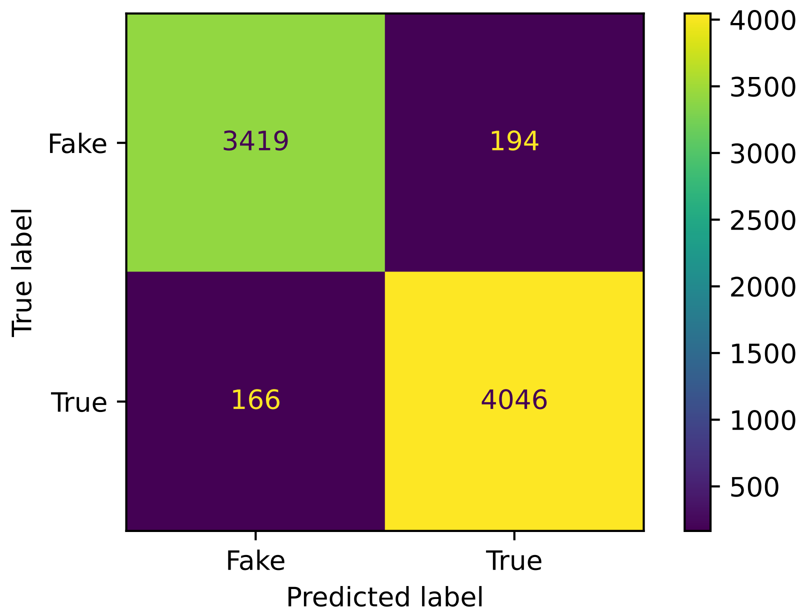

Figure 4: Confusion matrix for Naive Bayes classifier on test set

```
precision    recall  f1-score   support
```

	precision	recall	f1-score	support
Fake	0.95	0.95	0.95	3613
True	0.95	0.96	0.96	4212
accuracy			0.95	7825
macro avg	0.95	0.95	0.95	7825
weighted avg	0.95	0.95	0.95	7825

The model classifies both true and fake news correctly most of the time, indicated by o a high true postive and true negative rate. Both false positve and false negative rates are low, since there are few misclassifications. Therefore, not only is the model accuracy high, but the model is also robust in correctly detecting both true and fake news, with few misclassifications.

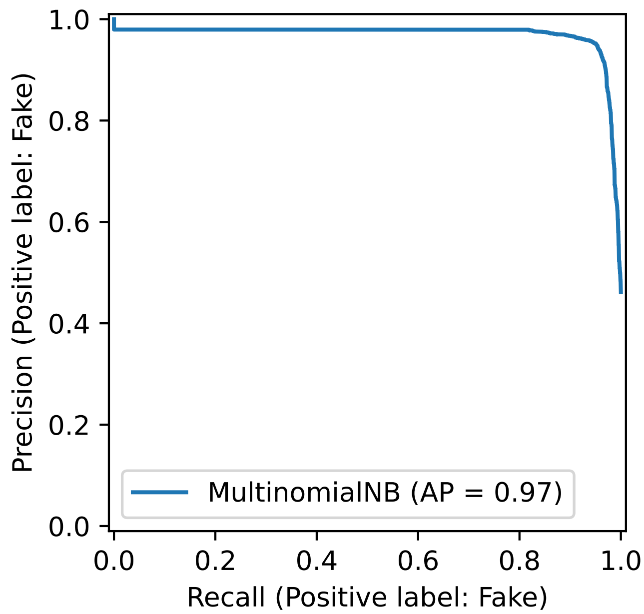

Figure 5: Precision-Recall curve for Naive Bayes classifier on test set

The AP score, as a summary of the PR curve, is 0.97. That fact that the AP score is very close to 1 suggests that our Naive Bayes model's performance is excellent.

The AUC here is 0.98, which is much higher than the baseline 0.5, further indicating the satisfactory performance of the Naive Bayes model.

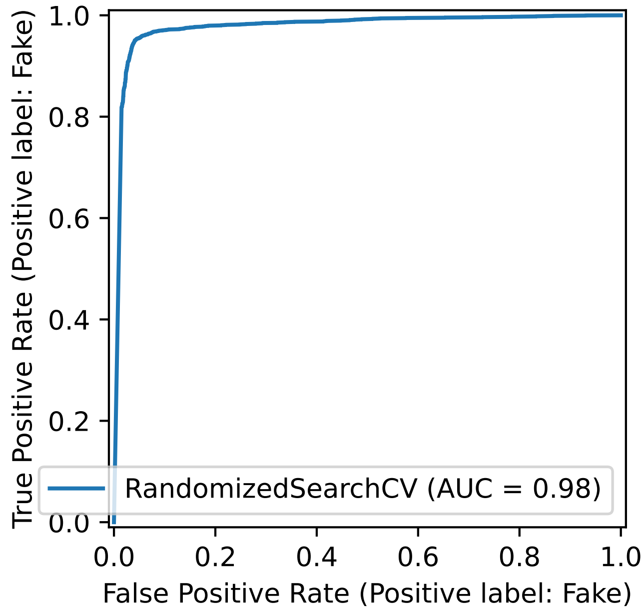

Figure 6: ROC Curve for Naive Bayes classifier on test set

6 Discussion

6.1 Result summary

The Naive Bayes model performs decently well due to high accuracy, high true positive rate, high true negative rate, low false postive rate, low false negative rate, high AP score and high AUC value. Apart from these exceptional evaluation metrics, this model also does not throw major concerns such as underfitting or overfitting problems. The training score (0.955), the best CV score (0.954), and test score (0.954) are consistently high and close to one another, so there is no noticeable underfitting or overfitting issue. To summarize, compared to the baseline model (dummy classifier), this Naive Bayes model exhibits major improvements, returning satisfactory results.

6.2 Discuss whether this is what we expected to find

Yes and No.

According to the nature of political news, it is expected that there are much more political news under the fake news category. However, in this dataset, there are only 2% more political news in fake news compared to true news. This marginal difference is not expected.

However, Naive Bayes model generally yields high performance despite its simplicity. So, using `MultinomialNB` with hyperparameter tuning gives an expected high prediction performance. Moreover, the original counts of words are used instead of the binary information presence/absence, so there is no loss of information in modelling. Therefore, the satisfactory result is as expected.

6.3 Discuss the impacts of these findings

This Naive Bayes model could potentially be deployed and used for news classification, which helps news readers identify fake news and avoid accepting false information. Secondly, this news classification model could also assist news agents screen and select only the true news to publish, avoiding losing reputation for articulating fake news. Lastly, this model could inspire further improvements of news classification models, acting as a good starting point for future model building.

6.4 Discuss what future questions could this lead to

Specific to this particular model, two future questions could be asked. First, the percentage of counts of political news is only marginally different between true and fake news. So, it is worth exploring whether dropping the `subject` feature will increase or decrease the model performance. This can be done by comparing the two models with and without the `subject` feature. Second, what other possible features can be used or engineered to further improve the model performance? This can be explored by asking for suggestions from human experts with domain knowledge. So far only the Naive Bayes model is used, which is a simple and naive model. Not restricted to this model anymore, there are other questions that can be raised. For example, are there any other more advanced models that can deal with integer count data and have more satisfying results?

6.5 Limitations of analysis

All news are classified as political and non-political in the analysis, which is crude and leaves space for improvement. More detailed `subject` categories can be applied to improve model performance. Besides that, all the collected data we have do not cover all types of news in the world. The topics included in the dataset are rather quite limited. So, this model could do well when seeing similar news, but could fail when given a totally unfamiliar news topic. So enriching the training data is essential to deploy the model in the wild.

7 Conclusion

The machine learning approach using Naive Bayes classification shows promising results for fake news detection. The model's high accuracy and balanced performance across both classes suggest it could be a valuable tool in combating misinformation. However, the model's performance may be limited to the types of news articles present in the training data, and further validation on diverse news sources would be beneficial for real-world deployment.

References

Allcott, Hunt, and Matthew Gentzkow. 2017. "Social Media and Fake News in the 2016 Election." *Journal of Economic Perspectives* 31 (2): 211–36. https://doi.org/10.1257/jep.31.2.211.

Bilodeau, Alexandre, and Farah Khalid. 2024. "Survey Series on People and Their Communities: Online Misinformation Concerns Among Canadians." *Statistics Canada*. https://www150.statcan.gc.ca/n1/pub/45-28-0001/2024001/article/00001-eng.htm.

Bisaillon, Clément. 2020. "Fake and Real News Dataset." https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset.

Service Canada. 2024. "Fake News and Misinformation." https://www.canada.ca/en/services/jobs/training/initiatives/skills-boost/digital-literacy/fake-news.html.